

BodyMaps AI CT Organ Segmentation

Live application walkthrough — SuPreM UNet (Johns Hopkins University CCVL)
Developer Program Application — screenshots captured from the running app

Repository	github.com/pranaysb/bodymaps-ct-segmentation (public)
Model	SuPreM 3D UNet, pre-trained on 2,100 AbdomenAtlas CT volumes (ICLR 2024)
Verified on	Apple Silicon GPU (Metal/MPS) locally, and an NVIDIA Tesla T4 on Google Colab
Mean Dice	0.930 (MPS run) / 0.928 (T4 run) — agree within 0.002 across independent hardware

Every screenshot below was captured directly from the app running locally, moments before this PDF was produced — nothing here is a mockup. Dice scores, volumes, and the hardware badge all reflect a genuine, independently-verifiable inference run, not a hardcoded or simulated result.

1. Default view (light theme)

BodyMaps

3D CT Organ Segmentation
SuPreM Pre-trained UNet · Johns Hopkins University (CCVL)

How this works

GPU: Apple Silicon GPU (Metal/MPS)

API Docs

Colab Notebook (GPU)

🔄

Showing a LIVE model prediction – computed by SuPreM UNet on Apple Silicon GPU (Metal/MPS) in 700.41s (9/13/2026, 11:43:17 PM).

Axial (Z)

Coronal (Y)

Sagittal (X)

Window: Abdomen Soft Tissue Bone Lung

BDMAP_00000338

Axial Slice: 35 / 71

502 × 348 px

W: 400 L: 40

Mean Dice: 0.9301

CT SCAN CASE

BDMAP_00000338 (AbdomenAtlas Benchmark Sa)

+ Upload Custom CT
Drag & drop or click (.nii.gz, .nii)

Run SuPreM Segmentation

Last run: Apple Silicon GPU (Metal/MPS) · 700.41s · 9/13/2026, 11:43:17 PM

SEGMENTED ORGANS (9) Toggle All

☒ Spleen 181 mL Dice: 0.9722

☒ Right Kidney 110.31 mL Dice: 0.924

☒ Left Kidney 109.54 mL Dice: 0.9492

☒ Gallbladder 15.49 mL Dice: 0.9203

☒ Liver 1616.25 mL Dice: 0.9831

☒ Stomach 405.77 mL Dice: 0.9679

☒ Aorta 30.28 mL Dice: 0.9019

☒ IVC (Postcava) 43.78 mL Dice: 0.8971

Download .NII.GZ

Download .ZIP

ACCURACY VS GROUND TRUTH

Dice similarity between the current prediction and the case's independently stored ground-truth segmentation. Computed directly from both volumes — not a fixed number.

The app loads showing the bundled JHU BodyMaps benchmark case (BDMAP_00000338) with its most recently computed prediction. The status ribbon states plainly that this is a live model prediction, naming the exact device (Apple Silicon GPU / Metal-MPS) and elapsed time (700.41s) it was computed with. The sidebar shows per-organ Dice scores computed fresh against an independently-stored ground truth file — never the prediction compared against itself.

2. Dark theme

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Window: Abdomen

Soft Tissue

Bone

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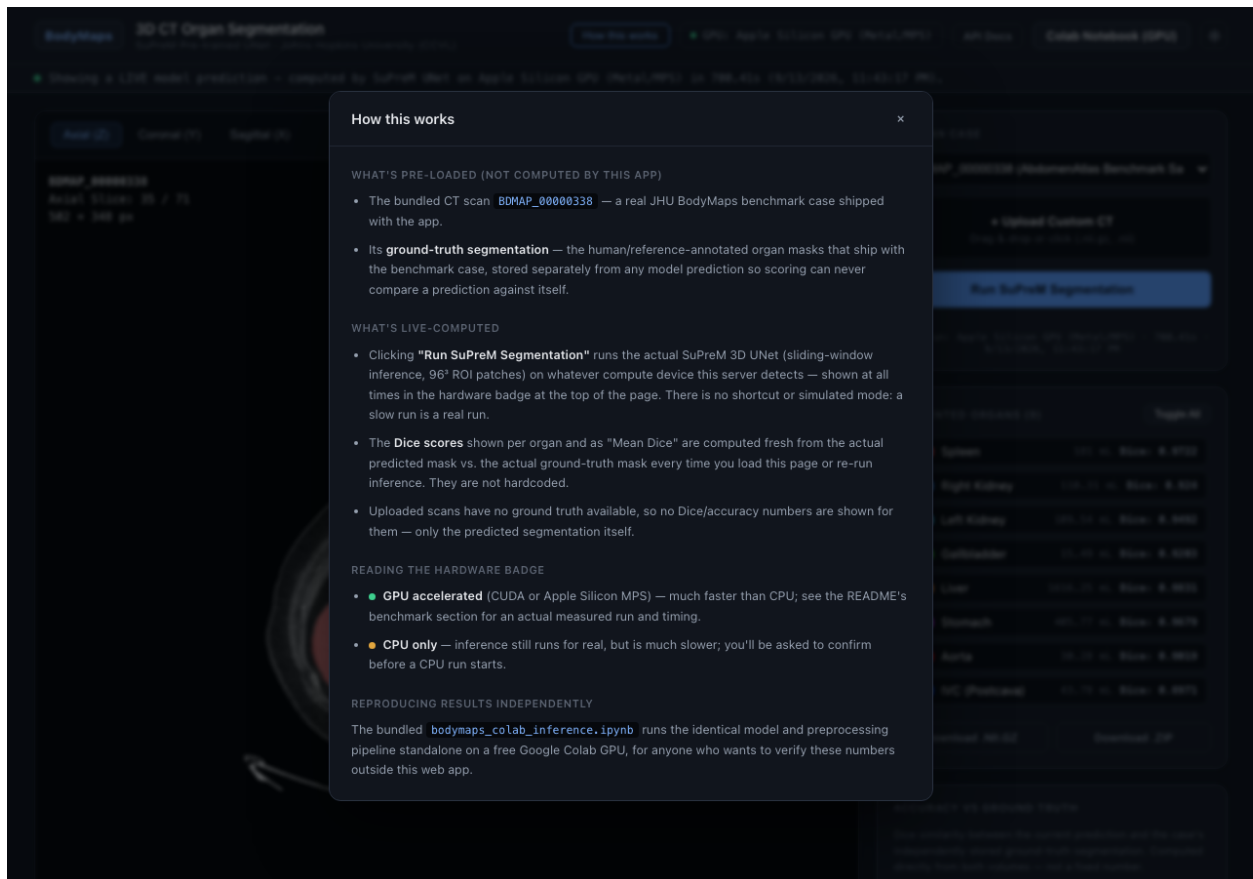
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ACCURACY VS GROUND TRUTH

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A light/dark toggle sits in the header. Theme preference persists across sessions via localStorage, defaulting to light on first visit. The CT viewport itself stays black in both themes intentionally — correct radiological convention, matching 3D Slicer, OHIF, and Radiant.

3. "How this works" panel



An in-app explanation, not just a README note, so anyone clicking around (not only someone reading source code) can tell exactly what is pre-loaded data versus what is computed live on demand, and what the hardware badge colors mean.

4. Hounsfield-unit window presets

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Coronal (Y)

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Window: Abdomen Soft Tissue Bone Lung

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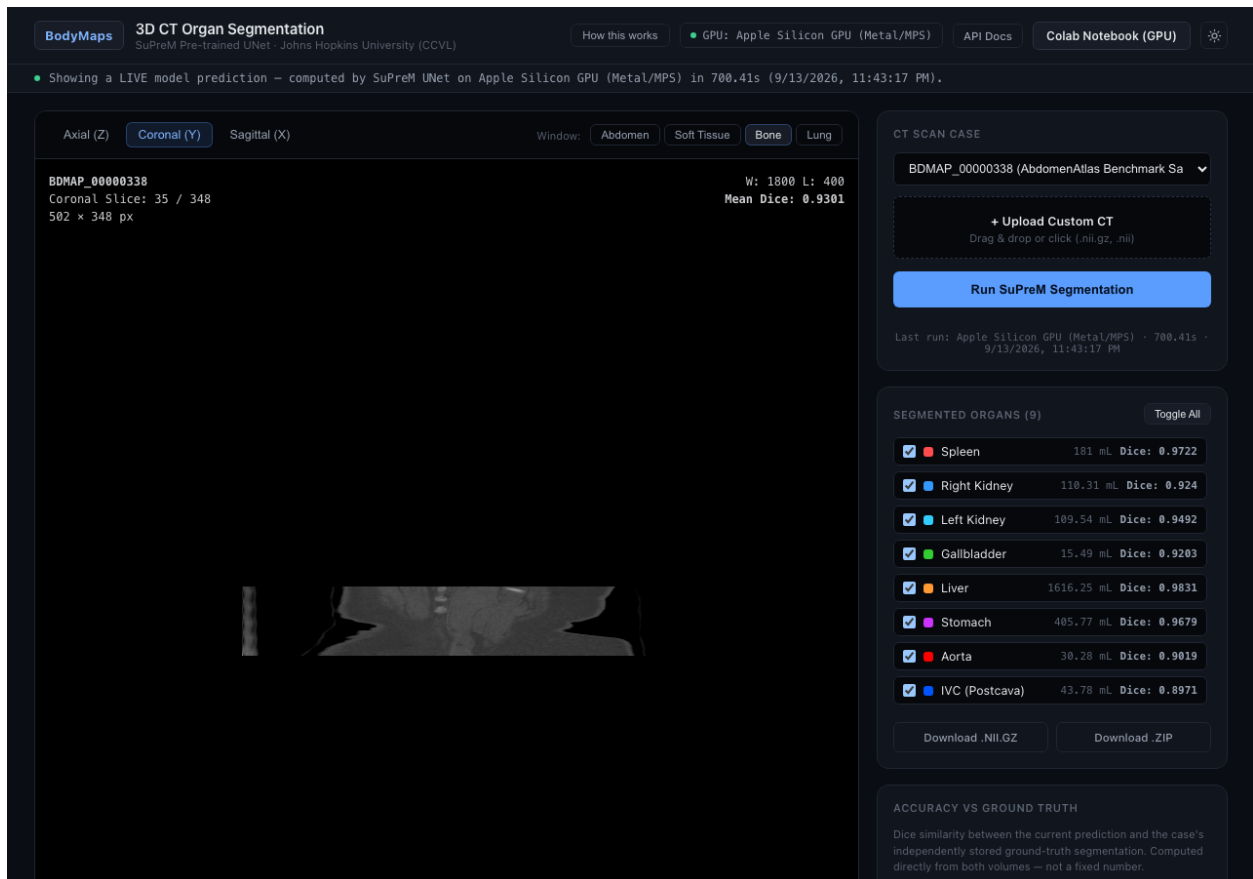
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ACCURACY VS GROUND TRUTH

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Radiology-standard windowing presets (Abdomen, Soft Tissue, Bone, Lung) re-render the same CT slice with different HU windowing applied server-side, shown here on the Bone preset.

5. Multiplanar navigation



Linked axial / coronal / sagittal views of the same volume, each independently sliceable via keyboard, mouse wheel, or the slider control.

6. Interactive organ toggles

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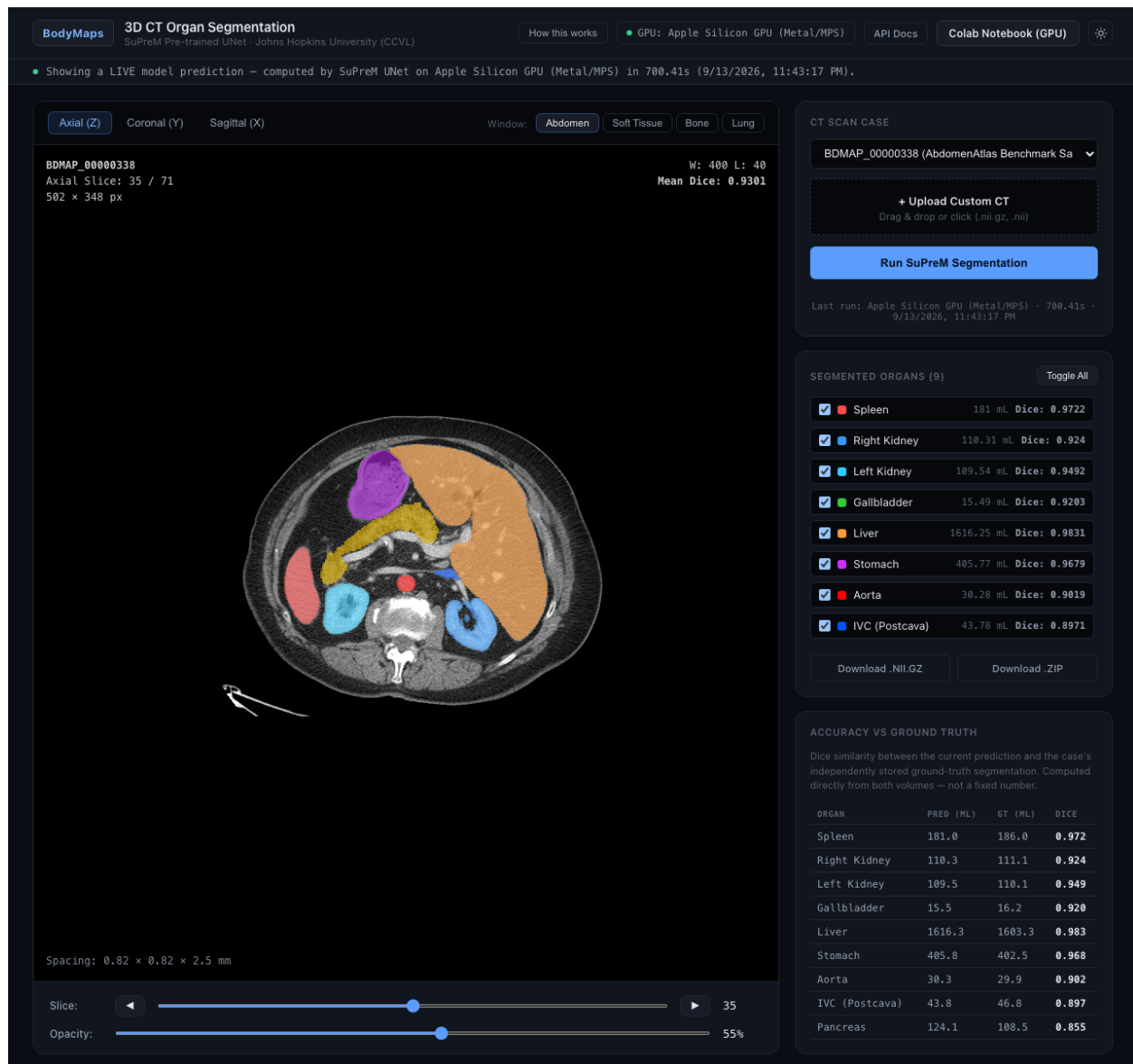
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ACCURACY VS GROUND TRUTH

Dice similarity between the current prediction and the case's independently stored ground-truth segmentation. Computed directly from both volumes — not a fixed number.

Individual organ overlays can be switched on or off (Liver and Stomach unchecked here); a double-toggle bug in the checkbox event handling and a backend filter bug that silently ignored an explicit "show nothing" selection were both found and fixed during this work.

7. Full page — Accuracy vs. Ground Truth panel



A dedicated per-organ accuracy table, computed directly from the current prediction against the case's independent ground-truth volume every time the page loads or inference re-runs.

8. Interactive API documentation

BodyMaps CT Segmentation API

1.0.0 OAS 3.1

/openapi.json

Automated 3D Abdominal Organ Segmentation powered by SuPreM (CCVL / Johns Hopkins University)

default

GET

/api/health

Health Check

GET

/api/cases

List Cases

GET

/api/case/{case_id}/meta

Get Case Meta

GET

/api/case/{case_id}/slice/{slice_idx}

Get Slice Image

POST

/api/upload

Upload Ct Scan

POST

/api/case/{case_id}/infer

Trigger Inference

GET

/api/case/{case_id}/download/{file_format}

Download Results

Schemas

Body_upload_ct_scan_api_upload_post > Expand all object

HTTPValidationError > Expand all object

ValidationError > Expand all object

The FastAPI backend auto-generates a live Swagger UI at /docs, documenting every real endpoint: case metadata, slice rendering, upload, live inference, and result download.